

Week 7 – Self-Study Tasks

Validation

I reviewed my Week 6 NLP work and my Group 4 project.

In the NLP task, I classified course messages using CountVectorizer, TF-IDF, Logistic Regression and Naive Bayes. I compared the models and also tested bigrams. My best result was **86% accuracy with CountVectorizer + bigrams + Logistic Regression**. I also checked the wrong predictions and found **40 duplicate messages**, which is an important limitation because similar messages can appear in both training and test data.

For my Group 4 project, I predicted whether a student's second quiz attempt would improve. I prepared **42 matched cases from 27 students**. I compared a baseline, Logistic Regression and Decision Tree. The results were **73%, 82% and 73%**. I then used a student-grouped split to make sure the same student was not in both training and test data. Logistic Regression reached **89% (8/9 correct)** with this split.

Visualization and Interpretation

I checked the target distribution, compared the model results and plotted the relationship between the first attempt score and improvement. There were **32 improved and 10 non-improved cases**.

The clearest pattern was that students with lower first scores improved more often. I also checked the Logistic Regression coefficients. The first attempt score had the strongest effect (**-1.29**), while week (**-0.38**) and duration (**-0.06**) had much smaller effects.

I also checked the wrong prediction from the grouped model. The student scored **32% first and 20% second**, but the model predicted improvement.

Method and Limitations

I used classification because my target was **improved / not improved**. I only used information available before the second attempt to avoid data leakage.

The main limitation is the small dataset. My definition of success also creates a ceiling effect because students with high first scores have less room to improve. For example, **100% → 100% is counted as no improvement**.

The dataset is synthetic. With real student data, privacy would be important, and model results should not be used to label a student's ability or motivation.

Conclusion

From my analysis, Logistic Regression performed better than the baseline and Decision Tree. The first attempt score was the strongest predictor of improvement.

However, I would not say that a low first score causes improvement. The dataset is small, and the result is affected by how I defined success. The analysis shows a pattern in this dataset, but it should be interpreted carefully.